微分熵 $h(X) = -\int f(x)\log f(x)\,dx$

[例] $X \sim U[0,a]$ $\quad h(X) = -\int \frac{1}{a}\log\frac{1}{a}\,dx = \log a$

Gaussian $X \sim N(0,\sigma^2)$ $\quad h(X) = \frac{1}{2}\log 2\pi e\sigma^2$

Pf: $\phi(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$

$$h(X) = -\int \phi\log\phi = -\int \phi\left[-\log\sqrt{2\pi}\sigma - \frac{x^2}{2\sigma^2}\log e\right]dx$$

$$= \log\sqrt{2\pi}\sigma\int\phi(x)\,dx + \int\phi(x)\frac{x^2}{2\sigma^2}\,dx\log e$$

给定 μ, σ^2 后，$f(X)$ 取 Gaussian 分布时 $h(X)$ 最大

Pf: $h(f) - h(\phi) = \int \phi\log\phi - f\log f = \int f\log\frac{\phi}{f} \le \log\int f\cdot\frac{\phi}{f} = 0$

Gaussian Channel

$Y = X + Z$ $\quad Z \sim N(0,N)$ $\quad$ AWGN

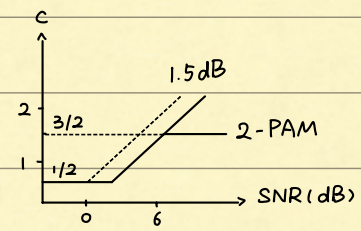
Capacity $C = \max\limits_{E(X^2)\le P} I(X;Y) = \frac{1}{2}\log\left(1+\frac{P}{N}\right)$

$$\begin{tikzcd} & Z \arrow[d] & \\ X \arrow[r] & \oplus \arrow[r] & Y \end{tikzcd}$$

Pf: $I(X;Y) = h(Y) - h(Y|X)$

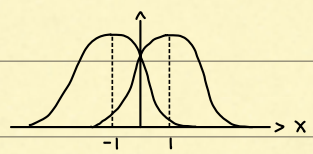
$h(Y|X) = h(Z) = \frac{1}{2}\log 2\pi e N$

$Var(Y) = P+N$, $h(Y) \le \frac{1}{2}\log 2\pi e(P+N)$



$R < C = \frac{1}{2}\log(1+SNR)$, $\quad SNR > 2^{2R}-1$ $\quad$ 6dB / bit

$P_e = Q(\sqrt{P/N})$



Gaussian Channel Capacity

$$C = \frac{1}{2}\log\left(1+\frac{P}{N}\right)$$

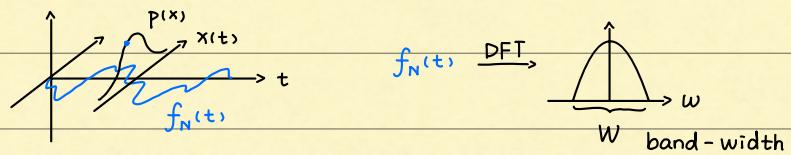
只考虑一个 symbol，实际每秒会有多个采样

Band-limited Gaussian Channel Capacity

$$C = W\log\left(1+\frac{P}{N_0 W}\right)$$

Pf：每秒约束总功率为 P

Noise power $\frac{1}{2}N_0$，每秒 2W 采样，$N = \frac{1}{2}N_0 \cdot 2W$



故 $f_N(t)$ 每秒至少 2W 个 sample

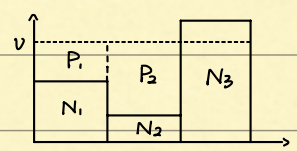
而后 $C = 2W \cdot \frac{1}{2}\log\left(1+\frac{P}{\frac{1}{2}N_0 \cdot 2W}\right) = W\log\left(1+\frac{P}{N_0 W}\right)$

Parallel Gaussian Channels

Capacity $C = \max\limits_{\sum E(X_i^2)\le P} I(X_1\cdots X_k; Y_1\cdots Y_k) = \max\limits_{\sum P_i \le P} \sum \frac{1}{2}\log\left(1+\frac{P_i}{N_i}\right)$

Pf：$I(X;Y) = h(Y_1\cdots Y_k) - h(Y_1\cdots Y_k \mid X_1\cdots X_k)$

$$= \underbrace{h(Y_1\cdots Y_k)}_{\le \sum h(Y_i)} - \underbrace{h(Z_1\cdots Z_k)}_{= \sum h(Z_i)} \le \sum \frac{1}{2}\log\left(1+\frac{P_i}{N_i}\right)$$

$\max \sum \frac{1}{2}\log\left(1+\frac{P_i}{N_i}\right)$ s.t. $\sum P_i \le P$，$P_i \ge 0$，KKT

$P_i = (\nu - N_i)^+$，Waterfilling